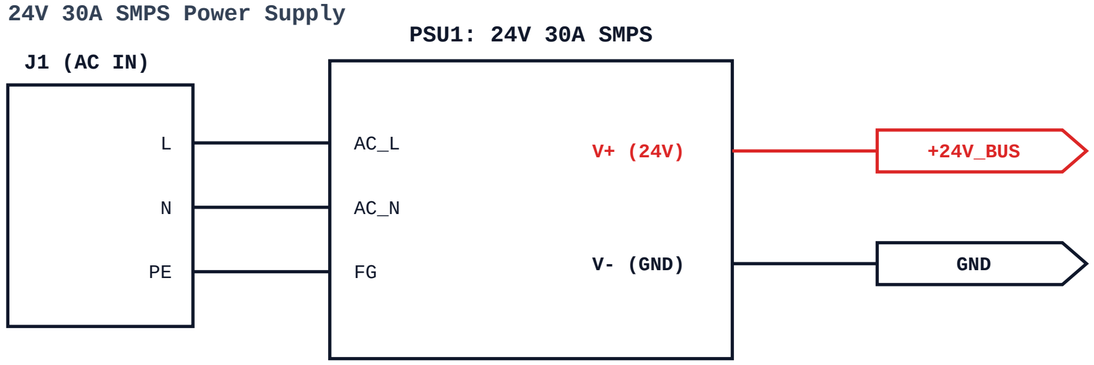
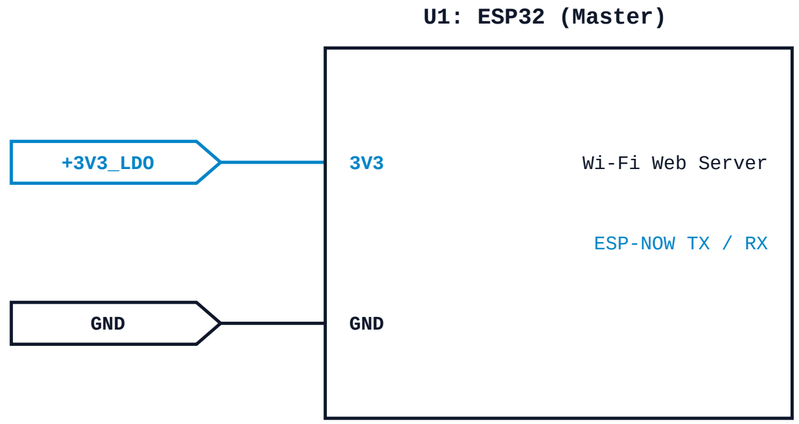
Sheet 1: Power Supply, Buck-LDO Regulation & Master ESP32

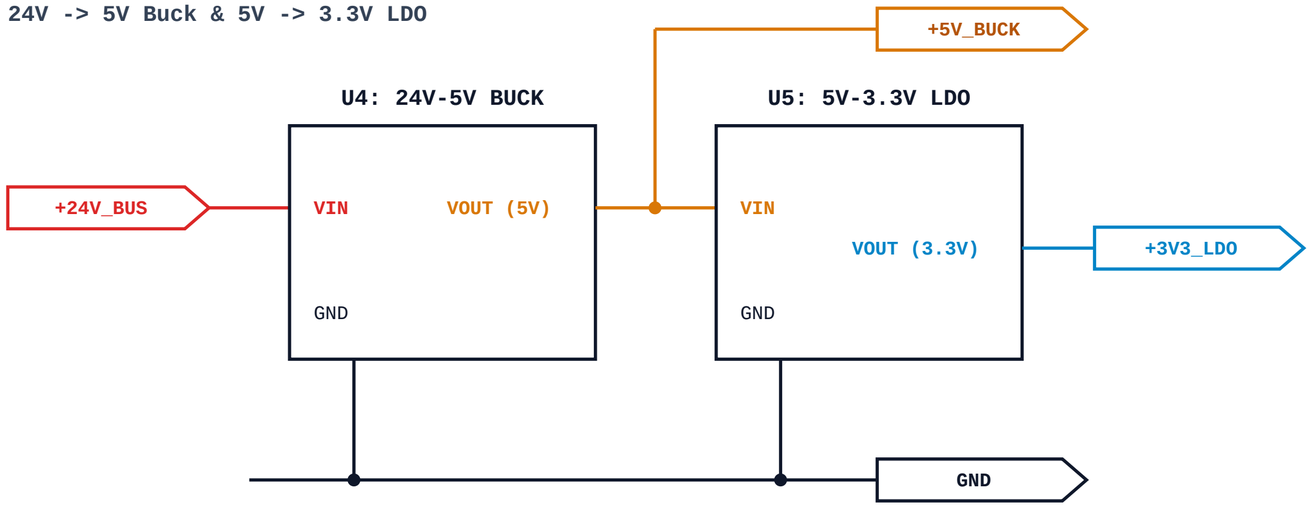
24V 30A SMPS Power Supply



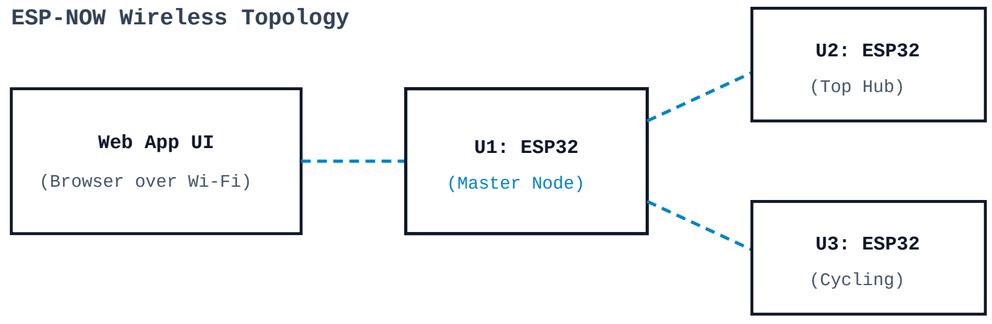
Master ESP32 (Web Server & ESP-NOW)



24V -> 5V Buck & 5V -> 3.3V LDO

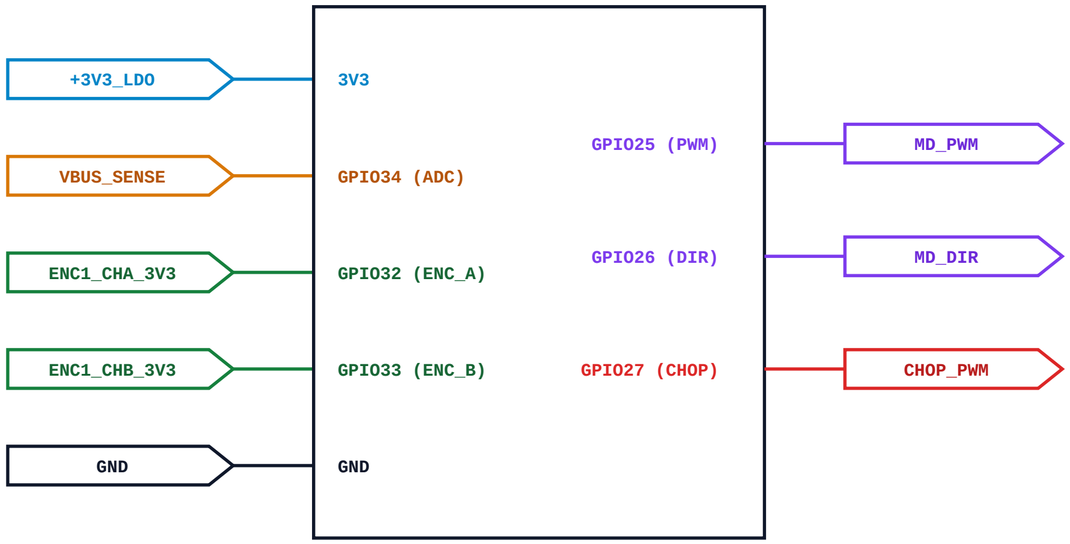


ESP-NOW Wireless Topology

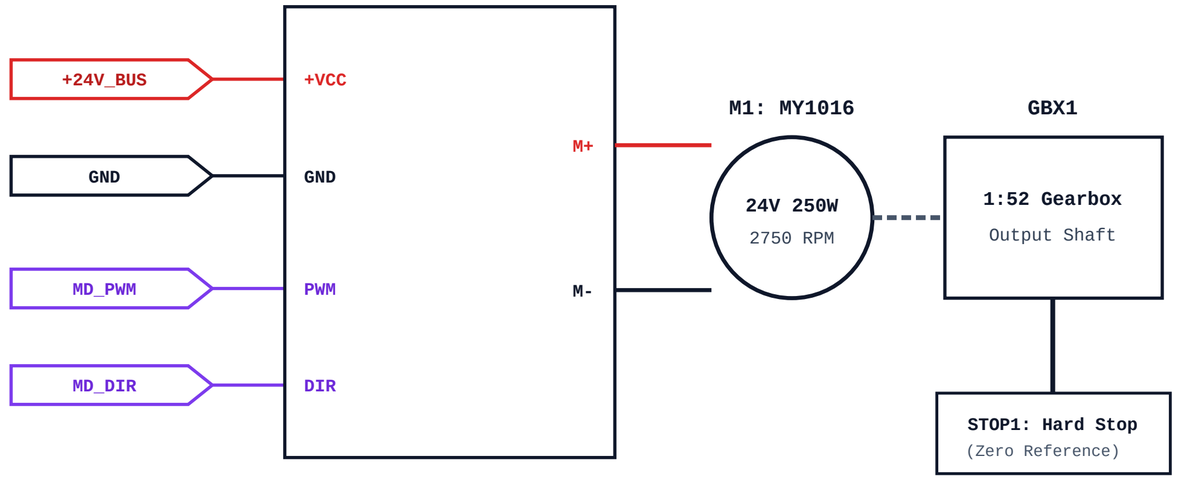


Sheet 2: Top Hub Motor Control, Brake Chopper & Optical Encoder

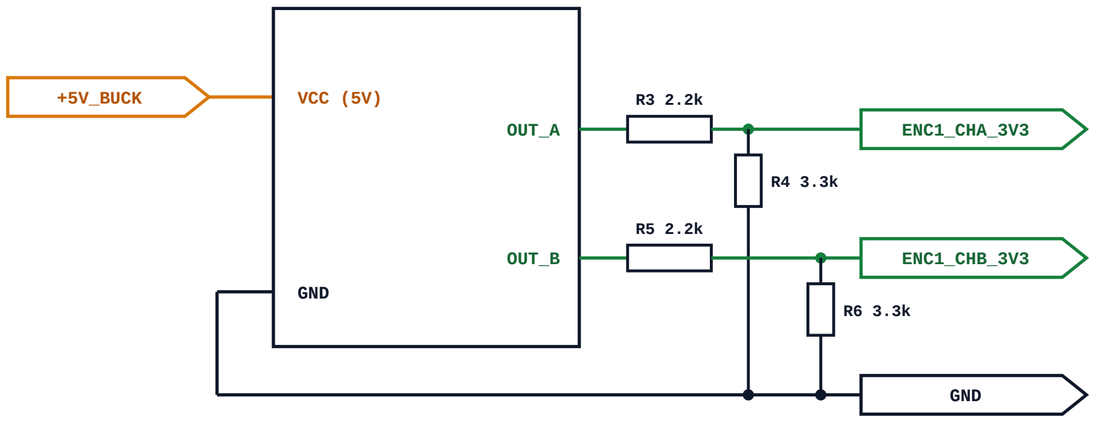
U2: ESP32 (Top Hub)



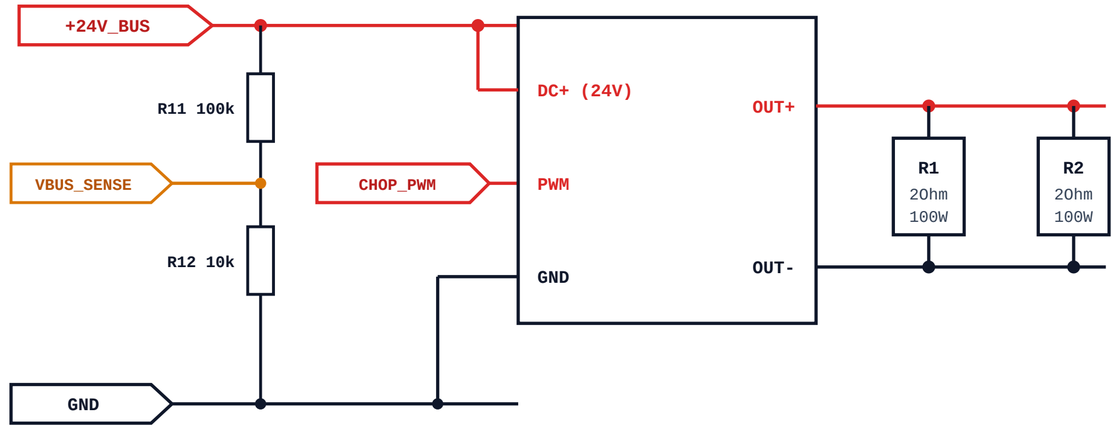
U201: Rhino MD20A



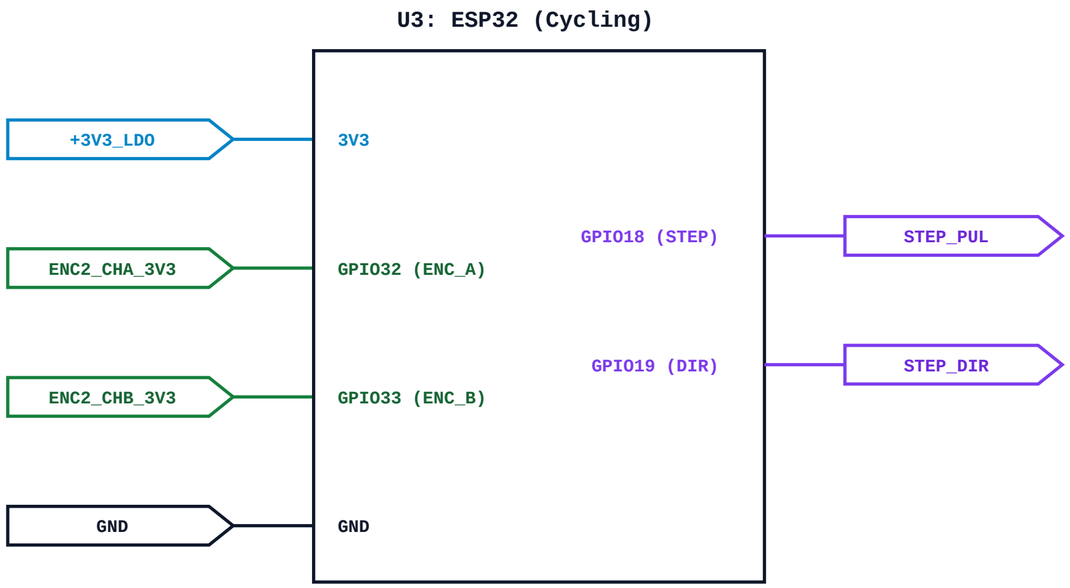
ENC1: 360 PPR

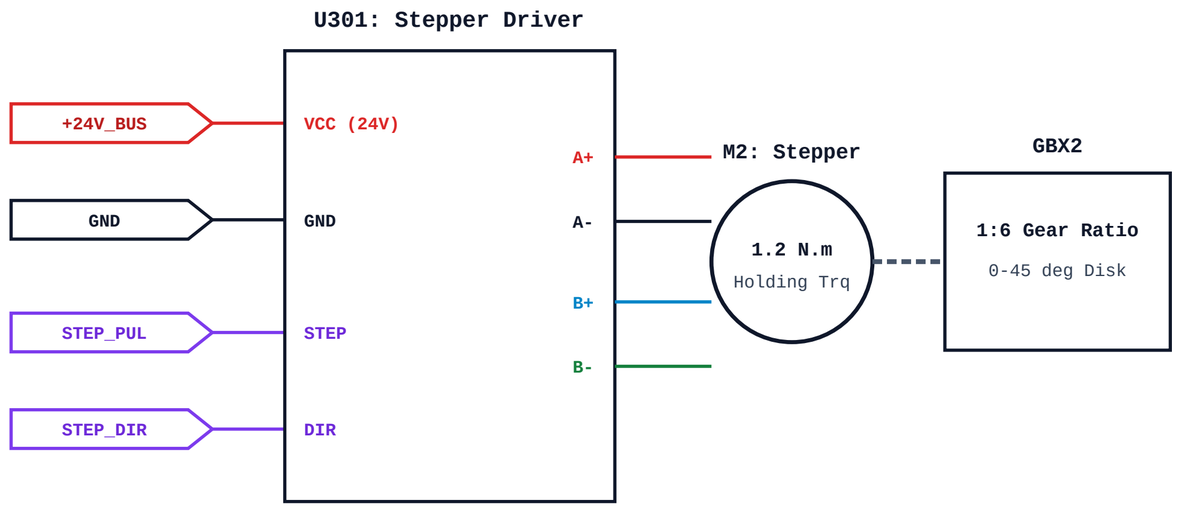


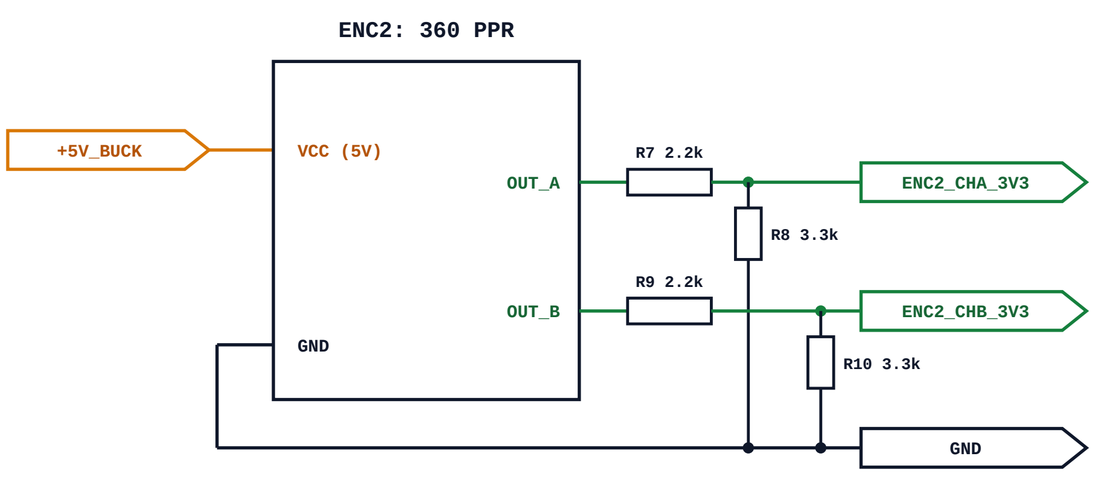
U202: 30A MOSFET

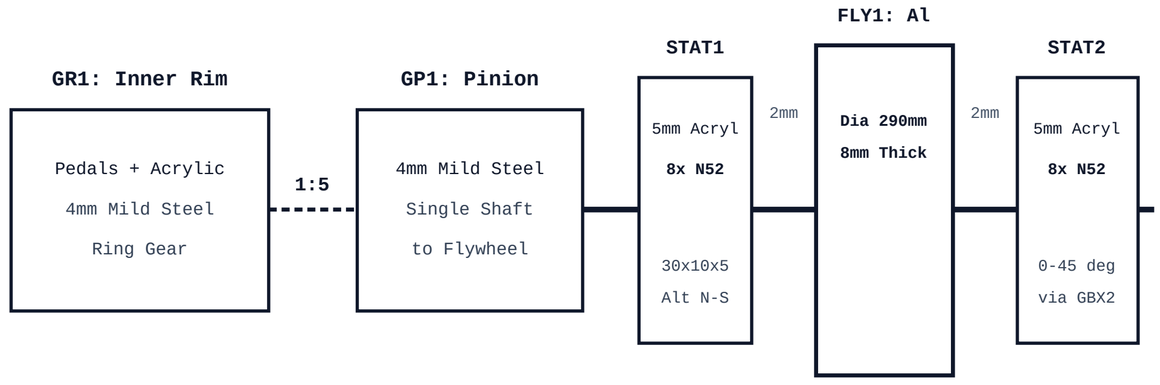


Sheet 3: Bottom Cycling Eddy Current Brake, Stepper & Flywheel









Sheet 4: Bill of Materials (BOM)

REF DES	COMPONENT	QTY	SPECIFICATION / DESCRIPTION
PSU1	24V 30A SMPS Power Supply	1	24V DC, 30A (720W) Switching Power Supply (Sheet 1)
U4	24V-to-5V Buck Converter	1	Steps down 24V bus to 5V for Encoders & LDO input (Sheet 1)
U5	5V-to-3.3V LDO Regulator	1	Linear regulator converting 5V to 3.3V for ESP32s (Sheet 1)
U1, U2, U3	ESP32 Microcontroller	3	U1: Master Web Server, U2: Top Hub, U3: Cycling (ESP-NOW)
U201	Rhino MD20Amp Motor Driver	1	6V-30V, 20A Single-Channel DC Motor Driver (Sheet 2)
M1	E-Bike DC Motor MY1016	1	24V DC, 250W, 2750 RPM Permanent Magnet DC Motor (Sheet 2)
GBX1, STOP1	1:52 Gearbox & Hard Stop	1	1:52 reduction gearbox on M1 + mechanical zero stop (Sheet 2)
U202	30A Heatbed MOSFET Module	1	3D Printer 30A MOSFET module for PWM brake chopper (Sheet 2)
R1, R2	2 Ohm 100W Braking Resistors	2	Ceramic aluminum-shell resistors wired in parallel (Sheet 2)
ENC1, ENC2	360 PPR Optical Encoder	2	2-phase incremental rotary encoders, 5V supply (Sheets 2, 3)
R3 - R10	Encoder Level-Shift Resistors	8	4x 2.2kOhm + 4x 3.3kOhm (5V to 3.3V divider for ENC1, ENC2)
R11, R12	Bus Voltage Sense Resistors	2	100kOhm & 10kOhm (24V to 3.3V divider for brake chopper ADC)
U301, M2	1.2 N.m Stepper & Driver	1	1.2 N.m holding torque stepper + driver + 1:6 gear GBX2 (Sheet 3)
GR1, GP1	1:5 Mild Steel Gear Set	1	4mm mild steel inner rim gear (on pedal acrylic) & pinion (1:5)
FLY1	Aluminum Flywheel Disk	1	290mm diameter, 8mm thick solid aluminum disk (Sheet 3)
STAT1, STAT2	Symmetrical N52 Magnet Disks	2	5mm acrylic sheets, 2mm air gap, 8x (30x10x5mm) N52 magnets/side